

Week 4 · Lesson 2 — Vite Build Tooling

Time: ~3–4h · **Goal:** Understand what a "build step" is and use Vite to develop fast and produce an optimized production bundle.

Why this matters for EVB-01

Browsers can't run JSX or `import` your component files directly. **Vite** does two jobs: (1) a fast dev server with hot-reload while you code, and (2) a `build` that bundles + minifies your React app into a few small static files. Those built files are what you'll gzip and flash onto the device — so Vite sits right in the middle of your deploy pipeline.

Core concepts

- **Scaffold a project:**

```
bash npm create vite@latest evb01-ui -- --template react cd evb01-ui && npm install
npm run dev # dev server with hot reload npm run build # produces the dist/ folder
npm run preview # serve the built dist/ locally to test it
(Vite needs a recent Node — 20.19+ or 22.12+.)
```

- **dist/ output:** an `index.html` plus an `assets/` folder containing hashed, minified files like `index-a1b2c3.js` and `index-d4e5f6.css`. The hashes are for browser cache-busting.
- **vite.config.js** is where you configure the build. **Critical for EVB-01:** set `base: './'` so the built `index.html` references assets with **relative** paths (`./assets/ ...`) — the device serves from its filesystem root, and absolute `/assets/` paths can break depending on how routes are registered.
- **Dev vs production:** during the week you'll mostly run `npm run dev` in your browser; only at deploy time do you `npm run build`.

Watch / read (in order)

- **[CORE]** Vite — *Getting Started*: <https://vite.dev/guide/> — official docs (note: `vitejs.dev` now redirects to `vite.dev`).
- **[CORE]** Vite — *Building for Production + Configuring Vite* (`base` option): <https://vite.dev/guide/build> and <https://vite.dev/config/>

- **[OPTIONAL]** Skim the `package.json` scripts Vite created so you know what `dev` / `build` / `preview` actually run.

Practice (do this) → push to git

1. Scaffold `evb01-ui` with the React template. Move your Week-4 Lesson-1 components into it. Run `npm run dev` and see hot reload work.
2. Set `base: './'` in `vite.config.js`. Run `npm run build`, then `npm run preview`, and confirm it still works.
3. Look inside `dist/` and note the file sizes (you'll compare against gzipped sizes next lesson).
4. Commit the project (with a `.gitignore` for `node_modules/`): `git commit -m "Week 4: Vite + React project, base set for embedded serving"` and push.

Before you move on, you should be able to...

- ☐ Scaffold, run, and build a Vite React app.
- ☐ Explain what `npm run build` produces and why files are hashed.
- ☐ Explain why `base: './'` matters for the device.